



Eduvos

Your Education. Your Future.

Project Deliverable 1 – Project Proposal

Faculty Name: Information Technology
Module Code: ITECA3-12
Module Name: Web Development and e-Commerce
Content Writer: Marc Kishinkwa
Copy Editor: Mr Kyle Keens
Submission Date:

Student Name	Sharleen Ngomakapile
Student Number	EDUV2958240
Project Title	Project Proposal
Submission Date	3/1/2025

Table of Contents

1.1 INTRODUCTION3

1.2 NEEDS/PROBLEMS5

1.3 GOALS/OBJECTIVES7

1.4 PROCEDURES/SCOPE OF WORK9

1.5 TIMETABLE11

1.6 CONCLUSION13

1.1 Introduction

The COVID-19 pandemic accelerated the shift towards online activities in South Africa, particularly in e-commerce with a report published in 2021 by Renub Research stating that South Africa's e-commerce market is on track to reaching an estimated value of US\$7.9 billion by 2027, growing at a compound annual growth rate of 9.8%. This expansion is pushed by the rise in internet penetration, increase in smartphone adoption and a shift in consumer preferences toward digital shopping experiences (GoGlobe, 2024). In 2020 alone more than 18 million South African's shopped online with an additional 6 million new users entering the market. This surge signals a major shift in the consumer behavior in South Africa, yet not everyone is benefiting from it. (Renub Research, 2021)

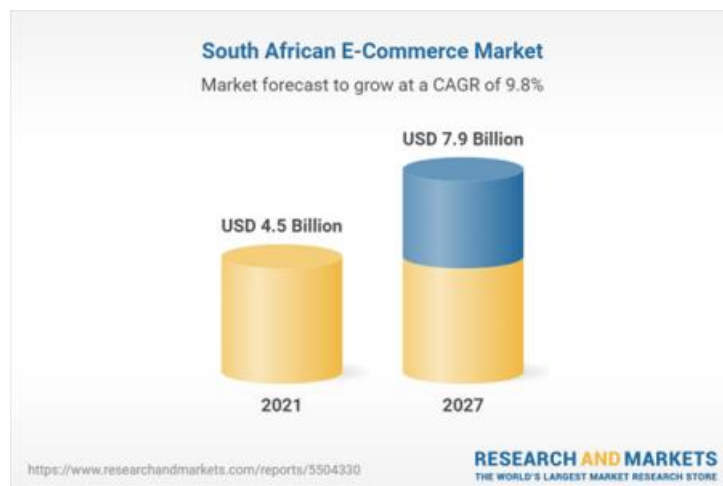


Figure 1: South African Market Forecast

Despite this expansion in the e-commerce market, the informal sector which employs over 5 million people is still struggling to transition online due to the lack of localized and accessible platforms (Alfers & Co-Ordinated, 2018). The market is dominated by foreign platforms that provide convenience but extract billions in potential revenue from the south African economy whilst failing to cater to the unique needs of local traders and buyers.. The lack of a secure dedicated C-2-C marketplace will continue to lock out informal traders from South Africa's digital growth. (GoGlobe, 2024)

This project aims to address these pressing issues by proposing a local C-2-C platform that empowers informal traders , keeps digital revenue within South Africa and boost economic growth by allowing smaller entrepreneurs to expand their field of reach and build sustainable businesses.

1.2 Needs/Problems

Limit in market reach for the informal economy

The informal sector in South Africa faces substantial challenges in accessing markets beyond their immediate local areas. Many smaller traders in the informal sector rely heavily on foot traffic in their physical locations which severely shrinks their customer reach. For example, street vendors selling handmade crafts in the township may only be able to attract customers from the surrounding neighbourhoods. This limited market access limits their business growth leading to a standstill in their incomes. Research from a Harvard Working Paper published in 2024, shows that a large majority of informal businesses in South Africa have not experienced significant revenue growth in the past five years with restricted market reach being one of the factors that contributed to this (Asmal, et al., 2024). The limit in market access has been a long-standing issue with attempts at government initiatives being launched to support informal traders however these efforts merely focused on providing physical market spaces or basic business training, but none have included digital first initiatives to connect these businesses with a wider market. (Arias, 2019)

Lack of a local C-2-C platforms

Currently South Africans who engage in C-2-C transactions rely mainly on international platforms such as Facebook Marketplace. However, although these platforms offer convenience they do not cater to the specific needs of the South African market. (International trade administration, 2021)

Although e-commerce has been expanding in South Africa since the early 2000s, the dominance of international platforms such as Facebook marketplace has persisted, and no comprehensive domestic replacement has yet emerged. There has been no major attempts to develop a all-inclusive domestic platform that addresses all the unique needs of the South African market. The dominance of foreign C-2-C platforms means that a significant amount of the revenue generated from e-commerce transactions leaves the country. This in turn reduces the amount of money that

can be reinvested into the local economy which affects economic growth and development. (International trade administration, 2021)

Concerns over trust and security

According to (Mofokeng, 2019) trust is another major concern in the C-2-C ecommerce space. Buyers are often times cautious of purchasing goods from unknown sellers due to the fear of the possibility of purchasing counterfeit products or that the seller may not deliver the products as promised (Mofokeng, 2019). Sellers on the other hand worry heavily about the possibility of the buyer not making payments or chargebacks. Trust and concern over security in C-2-C e-commerce has been around since the beginning of online trading in south Africa. As online facilitated transactions became more common so did the risks of cyber crimes and fraud increase over the past decade. South African banks and e-commerce platforms have implemented basic security measures such as two factor authentication (Mahir, 2024). However, such measures are not always enough to fully protect users. Trust and security concerns discourage the buyers and sellers from fully participating in C-2-C e-commerce. It also can destroy consumer confidence in online shopping in general which can have a ripple effect on the ecommerce models as consumers may become more and more cautious about making any online purchases (Vusani Netshirando, 2024). This in turn can cause the growth of the entire e-commerce industry in South Africa to slow down.

1.3 Goals/Objectives

The informal sector which is made up of township vendors and street sellers has been excluded from online trading due to three main barriers: limit in market reach, lack of localized C-2-C platforms, and trust issues. Developing a localized C-2-C platform will address these challenges. This project aims to create a user-friendly and secure e-commerce platform that empowers informal traders, keeps e-commerce revenue in South Africa and promotes economic growth. (International trade administration, 2021)

Specific, measurable, and achievable goals that will guide the development of this platform:

(1) Expanding Market Reach for Informal Traders

Informal traders in South Africa are currently restrained to local foot traffic, locking in their customer base to only physical proximity and setting a limit on their revenue potential. Introducing a digital first marketplace would launch these sellers into a broader more nationwide arena. By freeing informal vendors from the restrictions of township markets and street corners, the platform would unlock new revenue streams which enables small traders to compete in a digital economy that was previously beyond their reach. The result will not only be growth for individual sellers but a chain reaction for economic competitiveness across communities. (Fourie & Blaauw, 2023)

(2) Building a local C-2-C Platform

Develop and launch a local C-2-C platform that captures the South African C-2-C market share that is currently held by foreign platforms like Facebook Marketplace this will be achieved by integrating localized payment methods and providing multilingual support. Additionally, enabling smaller scale traders in the informal sector to increase their customer reach beyond their local areas and double their monthly revenue within the first months of using the platform. Market share and traders revenue will be monitored. Building a local C-2-C platform is beneficial as it retains e-commerce revenue for the local economic development

and empowers the informal sector which in turn reduces poverty and fosters local growth (Nastase, 2007).

(3) Building a Trustworthy and Secure Platform

The platform will implement an all inclusive trust and security framework. This includes identity verification for both buyers and sellers, secure payment gateway, and a conflict resolution system (BigCommerce , 2021). By the end of the first months the platform aims to achieve a high satisfaction rate among users regarding trust and security, which will be measured by user feedback surveys . The benefit of this goal is that it encourages more South Africans to participate in C-2-C e-commerce, with the confidence of knowing that their transactions are safe and secure. (Berger & Torres, 2024)

1.4 Procedures/Scope of Work

(1) Gather requirements

The initial phase in developing a C-2-C e-commerce platform for South Africa involves comprehensive requirement gathering (Builderfly, n.d.). This process includes reaching out to target users, specifically South African consumers engaged in C-2-C transactions. Questionnaires will be distributed across various regions and demographic groups to understand their preferences, such as the types of goods they typically buy and sell, and the features they expect from their ideal platform. For instance, some users might prioritize a secure payment gateway, while others may value an easy-to-use interface. Additionally, a detailed analysis of existing local and international C-2-C platforms will be conducted. This analysis will focus on user experience, features, and security measures to identify what works well and where the new platform can improve. (Builderfly, n.d.)

(2) Platform Development

For the front-end, HTML will be utilized to structure the various pages of the platform, including user registration, product listing, and checkout pages. CSS will be used to style these pages creating an aesthetically pleasing and intuitive interface. JavaScript will be implemented to add interactivity such as search functionality for products and dynamic updates to the shopping cart.

On the back end, PHP scripts will handle server-side operations, managing user authentication to restrict access to authorized users only. They will also manage products, including additions, updates and deletions. MySQL will be used to create and manage the database, storing user information, product details, and order history. The database will be designed with proper relationships to ensure data integrity and efficient retrieval.

(3) Testing

Testing plays a vital role in ensuring the platform functions as intended. Unit testing is employed to verify individual components, such as PHP user authentication and JavaScript form validation. Integration testing is then conducted to confirm how these components work together, examining processes like the flow from adding products to the cart through to order processing. Finally, user acceptance testing involves real South African users who provide valuable feedback on usability, functionality, and potential areas for improvement.

(4) Deployment of platform and on platform support

Once development and testing are complete, the platform will be deployed on a reliable hosting service. The selection of this service will be based on key factors including server performance, security measures, and cost-effectiveness. Following deployment, comprehensive user support will be made available. This includes a detailed FAQ section for common queries and a dedicated help section to address more complex issues. Additionally, thorough training will be provided to administrators to ensure they can manage the platform effectively.

1.5 Timetable

Deliverables	Description of Work	Start and End Dates
Deliverable 1	1. Researching domestic and international C-2-C e-commerce platforms in South Africa, primarily focusing on their impact on the informal economy and including suggestions for a new localised platform. 2. Developing a brief yet informative project proposal which includes the introduction, needs or problems, goals and objectives , scope or procedures and the conclusion	18-02-25 to 28-02-25
Deliverable 2	1. Designing a prototype for the e-commerce platform, creating diagrams like CRC, EERD, Context, DFD, and Use Case. 2. Partially developing the database-driven platform, writing sample code in HTML, CSS, JavaScript, PHP, and MySQL. Take screenshots of the platform and database tables.	15-03-25 to 06-04-25
Deliverable 3	Completing the e-commerce platform development, hosting on a free site, creating a user manual detailing technologies, features, and usage, and preparing a presentation for the lecturer.	13-04-25 to 20-04-25

Project Gantt Chart



1.6 Conclusion

The development of a dedicated C-2-C e-commerce platform in South Africa is necessary. Addressing the dominance of foreign C2C platforms and enhancing the informal economy is crucial for South Africa's economic growth and digital development (Etim, 2020). The project aims to create a user-friendly, secure, and local platform which is achievable and beneficial. It will increase economic activity, improve access to goods, and develop local e-commerce infrastructure.

References

- 1) Alfery, L. B. J. P. V. R. M. V. B. & Co-Ordinated, M. a. S. C., 2018. *IEJ*. [Online]
Available at: <http://kdpelmjpfafjppnhbloffcjpeomlnpah/https://iej.org.za/wp-content/uploads/2020/07/Stream-3-Policy-Brief-1-Informal-Economy-Sector.pdf>
[Accessed 20 February 2025].
- 2) Arias, J., 2019. *UPENN*. [Online]
Available at: https://pennur.upenn.edu/uploads/media/05_Arias.pdf
[Accessed 24 February 2025].
- 3) Asmal, Z., Bhorat, H., Lochmann, A. & Martin, L., 2024. *Harvard*. [Online]
Available at: https://growthlab.hks.harvard.edu/sites/projects.iq.harvard.edu/files/2024-04-wp-158-supply-side-economics-south-africa_0.pdf
[Accessed 24 February 2025].
- 4) Berger, I. & Torres, T., 2024. *Active Fence*. [Online]
Available at: <https://www.activefence.com/what-is-trust-and-safety/#:~:text=At%20its%20core%2C%20Trust,and%20misinformation.&text=create%20an%20environment%20where,and%20misinformation.&text=trust%20is%20key%20to,an,d%20misinformation.&text=and%20digital%20ethics%20help>
[Accessed 24 February 2025].
- 5) BigCommerce , 2021. *Big Commerce*. [Online]
Available at: <https://www.bigcommerce.com/articles/ecommerce/ecommerce-website-security/>
[Accessed 20 February 2025].
- 6) Builderfly, n.d. *Builder fly*. [Online]
Available at: <https://www.builderfly.com/what-are-the-main-challenges-of-ecommerce-in->

africa/#:~:text=Current%20challenges%20in%20ecommerce,infrastructure.%20...%0A

Coordination.&text=and%20Finance.%20...%0A Low%20degree,infrastructure.%20..

.%0A Coordination

- 7) Clifford Stein , Robert Drysdale, Kenneth Bogart, 2011. *Discrete Mathematics for computer scientists*. United states of America: Pearson Education.
- 8) Etim, E. a. D. O., 2020. The Informal Sector and Economic Growth of South Africa and Nigeria: A Comparative Systematic Review.. *Journal of Open Innovation: Technology, Market, and Complexity*, 6(4), p. 134.
- 9) Fourie, A. & Blaauw, D., 2023. *The conversation*. [Online]

[Accessed 24 February 2025].

Available at: <https://www.go-globe.com/e-commerce-south-africa/#:~:text=Localization%20ensures%20that%20E%2Dcommerce,social%20circles.&text=to%20resonate%20with%20South,social%20circles.&text=to%20consumers.%20A%20personalized,social%20circles.&text=a%20connection%20to>

Available at: <https://www.go-globe.com/e-commerce-south-africa/#:~:text=E%2Dcommerce%20has%20witnessed%20remarkable,the%20country.&>

ext=internet%20penetration%2C%20rising%20smartphone,the%20country.&text=transforming%20the%20retail%20landscape,the%20country.&text=vast%

[Accessed 25 February 2025].

- 12) International trade administration, 2021. *international trade administaration*. [Online]

Available at: <https://www.trade.gov/country-commercial-guides/south-africa-ecommerce#:~:text=South%20Africans%20purchase%20mostly,business%20use.&text=States%2C%20and%2014%20percent,business%20use.&text=in%20South%20Africa.%20If,business%20use.&text=times%20a%20year%20>

[Accessed 25 February 2025].

- 13) Mahir, M., 2024. *Linkedin*. [Online]

Available at: <https://www.linkedin.com/pulse/securing-e-commerce-platforms-africa-protecting-online-mehdi-mahir-icese#:~:text=Adding%20an%20extra%20layer,gain%20access.&text=helps%20to%20prevent%20unauthorized,gain%20access.&text=requires%20users%20to%20verify,gain%20a>

[Accessed 24 February 2025].

- 14) Mofokeng, T., 2019. *Easy chair*. [Online]

Available at: <https://easychair.org/publications/paper/5cf1/open>

[Accessed 24 February 2025].

- 15) Nastase, I. A., 2007. Contribution of the e-commerce to the economic development..

International Conference on Business Excellence.

- 16) Renub Research, 2021. *South Africa E-Commerce Market Size, Forecast 2022-2027, Industry Trends, Share, Growth, Impact of COVID-19, Opportunity Company Analysis*, s.l.: Renub Research.

- 17)Vusani Netshirando, A. K. a. M. W., 2024. Effects of cyber-security and trust issues on consumers' intention to use E-commerce systems in rural South Africa. *nternational Journal of Research in Business and Social Science*, 13(7), pp. 538-549.